



Degree Project in Applied and Computational Mathematics

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An Attempt to Synthesise Predictive Modelling with Optimal Transport for Resource Allocation in Large- Scale Construction

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Abstract

This report presents a novel approach for combining predictive methods, in particular, Gaussian Processes (GPs) with the Optimal Transport (OT) framework in large-scale construction settings handled by the Swedish Transport Administration (STA). The aim is to explore the possibility to forecast excavations and demands on multiple construction sites and formulate this as a mass allocation problem. The methods used are presented in a breadth-first style, providing an overview rather than a deep introduction. Given the large topic of OT, the major part of this thesis consists of the presentation of this and some of its measure theoretic prerequisites. First we develop a GP regression model to individually forecast excavated resource supply and landfill demand at a collection of construction sites, leveraging a feature engineered and generated dataset mimicking the structure of the real dataset in the STA's young data collection project ELSA. The model shows potential to predict quite well in comparison to prior fictively generated estimates made by the contractor and the STA – generated via variances multiplied with true function and some white noise added to it to include heteroscedastic estimates – while also providing uncertainty quantification. Secondly, we formulate the OT problem as an Unbalanced Optimal Transport (UOT) problem with entropic regularisation, allowing for so called mass destruction and teleportation. This entropic UOT is then solved using the Sinkhorn algorithm. The results shows the forecasted system can be modelled using OT and illustrates how supply and demand efficiently can be satisfied. Finally, implementing both predictive methods and OT as well as the combination to real data is recommended to the STA as the findings shows it can be a useful addition to their existing estimation methods. In particular, an exploration of extended methods such as combining predictive models, uncertainty quantification and robust optimisation in UOT is recommended for future work.

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List of Notations and Symbols

- $L^1(d\mu)$ The space of measurable functions f for which $\int |f|d\mu < \infty$.
- $P_p(\mathcal{X})$ The Wasserstein space of order p .
- $T_{\#}\mu$ A push forward measure of μ by the mapping T .
- $W_p(\mu, \nu)$ The Wasserstein distance of order p .
- $\Pi(\mu, \nu)$ The set of possible transport plans π .
- $\langle \mu, f \rangle_{\mathcal{X}}$ Short notation for the inner product $\int_{\mathcal{X}} f d\mu(x)$.
- $\mathcal{C}(\mathcal{X})$ The set of continuous functions on $\mathcal{X}$.
- $\mathcal{D}_{\varphi}(\mu \mid \nu)$ The divergence w.r.t. an entropy function φ .
- $\mathcal{B}(\mathcal{X})$ Borel σ -algebra of $\mathcal{X}$.
- $\mathcal{M}(\mathcal{X})$ The set of σ -additive functions $\mu : \mathcal{B}(\mathcal{X}) \rightarrow \mathbb{R}$.
- $\mathcal{P}(\mathcal{X})$ The set of probability measures on $\mathcal{X}$.
- $\nu \perp \mu$ The measures μ and ν are mutually singular.
- $\nu \ll \mu$ A measure ν is absolutely continuous w.r.t. a measure μ .
- $\pi(x, y)$ Transport plan of mass from $x \in \mathcal{X}$ to $y \in \mathcal{Y}$.
- φ'_{∞} The speed of growth at ∞ of an entropy function φ .
- $\varphi^*(p)$ The Legendre transform of an entropy function φ .
- $c(x, y)$ **or** $C(x, y)$ The cost of transporting a unit of mass from $x \in \mathcal{X}$ to $y \in \mathcal{Y}$ in continuous settings or discrete settings, respectively.
- $p_{\mathcal{X}}$ The projection map onto the space $\mathcal{X}$.

ACF and PACF AutoCorrelation Function and Partial AutoCorrelation Function, respectively.

ELSA EnergiLedningsSystem för Anläggningar (Energy management systems for facilities).

GHG Green House Gas.

GP Gaussian Process.

OT Optimal Transport.

STA Swedish Transport Administration.

UOT Unbalanced Optimal Transport.

Chapter 1

Introduction

The Swedish Transport Administration (STA) handles large projects all over Sweden. One section is the STA's Stora Projekt (Large Projects) where they handle projects with a turnover exceeding a billion SEK. For instance, these larger projects consist of highway or road connections and the development of railway systems such as the West link (Västlänken). In this project, the focus will lie on the potential applications of the theory presented here to the Cross-connection (Tvärförbindelsen) Södertörn, spanning between Botkyrka and Nynäshamn. It connects larger roads with the new larger harbour located in Nynäshamn that eventually is to replace the central harbour Frihamnen in Stockholm. In particular, the resources handled in the construction sites within the Cross-connection will be of interest here. These resources consists of mainly rock, soil, gravel, cement and reinforcement bars and are either being excavated on site or transported to a construction site from outside. The excavated resources that is not reused on-site are then transported out of the sites while resources used for landfills or constructions are transported into the sites.

Moreover, the significant amount of resource movements implies large environmental, social and economical consequences, including greenhouse gas emissions, road wear and disturbance of nearby public road traffic. Thus, there is a need to minimise the number of transports being made. In the challenge of minimising transportations and waste of resources, one should investigate which excavated resources that can be reused or recycled within the project area and, hence, do not need to be transported out of the projects.

For instance, rock is a likely resource to be excavated in Sweden and can also be used in paving and landfills or crushed into gravel for finer works; and perhaps even reused as aggregate in concrete mixing. Given the density of granite the number of transportations omitted would increase quite rapidly if it could be recycled within the projects and particularly if resources can be recycled inter project-wise. Thus, having some prior knowledge of the resource circulation or occurrence – i.e., when and in what quantities the resources and materials are being excavated or delivered – would enable better allocation of resources according to the demand and supply across project sites. This prior insight is, however, currently limited to empirical knowledge only, as far as we know, and not made within a theoretical framework at the STA. Thus, there is a need for additional estimation methods for forecasting the amount of reusable excavated materials within these projects as well as in projects nearby while also forecasting the demand of these materials within the considered construction sites, using a theoretical structure. This can provide the STA and the constructor with a decision base and potentially better control of transportation needs, facilitating planning for storage and more accurate transport scheduling in order to mitigate negative effects.

In this report, *Gaussian processes* (GP) will be introduced and implemented to estimate the occurrence of resources within major projects. GPs does nicely put empirical knowledge into a theoretical structure, allowing for more robust predictions. The estimates can likely be based on data collected in the STA’s new and ongoing project ELSA that collects activity data from the constructions sites, including the amount of transportations, quantities of materials needed and how much waste materials that is produced, to name a few [23]. However, due the recent inception of ELSA as well as confidentiality constraints, this report will use self-generated data designed to reflect the structure of the ELSA dataset.

Furthermore, when predictions have been made, the resource allocation problem persists. Here it is, however, not necessarily the number of transports that should be minimised; there are as mentioned other factors to consider such as environmental, social and economical consequences. Ideally, by quantifying some of these causal effects of transporting different resources one might be able to define a cost function capturing most of these negative consequences. By integrating this cost function into a system of resource flows one can define the total cost of transporting resources between construction sites satisfying supply and demand. Naturally, one aims to minimise this

total cost. This minimisation problem is called *Optimal Transport* (OT) and will be the main focus of this report. Hence, an introduction to the basics of OT as well as measure theoretic preliminaries utilised in it will be made. Moreover, the STA case study will serve as an example of a real world application of OT. Although, applications of OT spans many areas and disciplines of which a few will be briefly discussed.

Chapter 2

Introduction to Gaussian Processes

We will here introduce the method used in the case study for the STA. Gaussian Processes (GP) are used in this study, for which a description of how the data is interpreted is intended to work as a motivation of. While there exists alternative methods, particularly for capturing complex dependencies or error distributions that might not fully be captured by standard GP, the GP framework is a suitable starting point given the context of the data and the desire for quantifying uncertainties. Furthermore, a selection of kernel functions used in GPs will be discussed as well as some of their corresponding hyperparameters. Selecting kernels and hyperparameters properly is crucial for adapting the GP model to complex variations in the data, including kernel parameters and noise variance, as well as for finding a satisfactory bias variance trade-off.

2.1 Data Specification and Interpretation

Data will be considered as coming from a space $\mathcal{X}^p \times \mathcal{Y}$, often $\mathbb{R}^p \times \mathbb{R}$, and be of the form $\mathcal{D}_{TR} = \{(x_i, y_i)\}_{i=1}^n$ that are outcomes of some independent random pairs $\{(X_i, Y_i)\}_{i=1}^n$, where X_i is an explanatory vector in $\mathbb{R}^p$ and $Y_i \in \mathbb{R}$ its corresponding response variable. The collection of explanatory variables is then interpreted as the matrix $X = (X_1, \dots, X_n)^T \in \mathcal{X}^{n \times p}$ and the response vector as $Y = (Y_1, \dots, Y_n)^T \in \mathcal{Y}^n$. Moreover, we aim to perform a supervised

learning for prediction; the updated outcomes $X_* = x_* = (x_{n+1}, \dots, x_{n+k})^T$ will be collected together with X and Y to predict the response $Y_* = \hat{f}(X_*) + \varepsilon$, where ε follows some distribution that ideally can be transformed into a normally distributed error $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_k)$. Moreover, being able to quantify the uncertainties of the predictions with a Confidence Interval (CI), would be a valuable informative addition. Thus, we want to pick a method or possibly a combined approach providing the parameters to create a CI with.

The function $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ is the estimation of the true function capturing the behaviour of the observations. Furthermore, mutually independent and normally distributed error simplifies statistical inference but is, however, not necessarily the case which might imply that there are learning methods such as tree-based methods that will perform and capture variations better; or alternatively if one implements variational inference or Monte Carlo methods to track the true posterior distribution of these. One might also be able to model the dependencies between noise using an ARMA process as in [14].

In the practical setting of this report, the sample space $\mathcal{X}$ consists of both categorical and quantitative data while the target space $\mathcal{Y}$ is the real line representing the surplus or deficit of the material masses in each construction site. All together, the aim is to implement a supervised learning algorithm to estimate multiple regressors with functions $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$.

The collected data consists of activity data for each construction site such as deliveries of goods and occurrences of waste. We can interpret the response variable y as multivariate as well, since predictions of landfill demands, excavation surpluses as well as the resource going to depot, for each resource is to be made. Moreover, the explanatory variable x includes data regarding identification of the construction sites informing of what types of projects such as tunnels, roads or bicycle lanes, the contractor and their prior estimations of the total amount of excavated resources, the STA's prior estimations, the actual amount of excavations and the transportations of these. While it seems to be a natural time ordering of the data, we will not consider the time dynamics explicitly but rather capture seasonal patterns from categorically constructed features such as 'winter' or 'summer' from the month of the year and possible ground frost aggravating excavations. furthermore, it will be assumed the data is mutually independent.

The structure of the described data makes the task of predicting excavated and demanded resources for constructions sites suitable for GP Regression. In

fact, most assumptions made about the data are assumptions usually made when implementing GPs. However, since new data often appear during a project's lifetime it is possible online learning methods are more suitable in a practical implementation, hence, might therefore be of interest in the future.

2.2 Gaussian Process Regression

Gaussian Processes provide an excellent and flexible framework for regression tasks where uncertainty is to be quantified. It is a probabilistic, Bayesian and non-parametric method that can provide a prediction interval of a response vector Y given some prior data X, Y and some newly observed explanatory vector X_* . GPs can be defined, as it is done in [28, 9], as follows:

Definition 2.2.1. A Gaussian process on $\mathcal{X}$ is a collection of random functions $\{f(x)\}_{x \in \mathcal{X}}$ such that for any finite collection $(x_1, \dots, x_d)^T \in \mathcal{X}^d$ the random vector $(f(x_1), \dots, f(x_d))^T$ has a multivariate normal distribution.

Furthermore, the mean and covariance function will be written as $m(x) = \mathbb{E}[f(x)]$ and $K(x, x') = \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))]$, respectively, sometimes shortly m and K when not too ambiguous. A key property for a GP $f(X) \sim \mathcal{N}(m, K)$ that can be partitioned as

$$f(X) = \begin{pmatrix} f(X_1) \\ f(X_2) \end{pmatrix}, m = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}, K = \begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix}$$

is that the conditional is distributed as $f(X_1) \mid f(X_2) = f(x_2) \sim \mathcal{N}(m_{1|2}, K_{1|2})$, where $m_{1|2} = m_1 K_{12} K_{22}^{-1} (f(x_2) - m_2)$ and $K_{1|2} = K_{11} - K_{12} K_{22}^{-1} K_{21}$ [9].

When implementing GP regression we aim to estimate the predictive distribution of the conditional $Y_* \mid X_*, X, y$. Thus, the sought predictive distribution is tractable and known to have the form

$$Y_* \mid X_*, X, y \sim \mathcal{N}(m_*, K_*), \text{ where the predictive mean is } m_* = K(X_*, X)[K(X, X) + \sigma^2 I_n]^{-1} y \text{ and the predictive kernel is } K_* = K(X_*, X_*) + \sigma^2 I_k - (K(X_*, X)[K(X, X) + \sigma^2 I_n]^{-1} K(X, X_*)), \quad (2.1)$$

see e.g. [9, 28]. One prominent issue is the computationally expensive matrix inverse that is to be computed. This is, however, not a problem in the practical implementation of this project since the number of points are fairly

few and the dimension not too large. Moreover, implementation of Tensorflow’s GPflow on GPUs can be done, if necessary, significantly increasing computational speed.

2.3 Kernel functions

The kernel function plays, as mentioned, an important role in the predictions made with GPs, hence, it is reasonable with a brief introduction. Kernels are essential in GPs since they capture similarities between input points as well as prior assumptions about the true function being modelled, such as its smoothness [28].

Two common types of kernel functions used in practice are stationary and isotropic kernel functions. A *stationary* kernel function is a function of $\tau = x - x'$ while an *isotropic* kernel function depends on $r = |x - x'|$. Some common kernel functions used in practice include the Matérn and the Square Exponential (SE), also called the radial basis function. These kernels are, respectively, defined as

$$\begin{aligned} k_{\text{Matérn}}(r) &= \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}r}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}r}{\ell} \right), \\ k_{SE}(x, x') &= \sigma_f^2 \exp \left(-\frac{r^2}{2\ell^2} \right), \end{aligned} \tag{2.2}$$

where ν , ℓ and σ_f^2 are hyper parameters tuned to make the kernel function capture variations of the data, and $r = \|x - x'\|$ (or some other defined distance such as the Mahalanobis distance). The hyperparameters ν and ℓ are positive and control the smoothness and the length scale or decay of correlation, respectively. K_ν in the Matérn kernel is a modified Bessel function. The Matérn is quite special, and when the smoothness parameter ν is half an integer it becomes a product of a polynomial and exponential kernel as can be seen in [28]. Moreover, products and sums of kernels are again a kernel function. This allows for creation of a large variety of kernel functions that can be adapted to one’s specific task at hand.

There is much more to say about kernel functions, for instance, the Mercer’s theorem showing the possibility to represent kernel functions with their associated eigenfunctions and eigenvalues. Moreover, Williams and Rasmussen

[28] mentions the development of GPs much depends on the research and expressiveness of covariance functions, which the reader is referred to for a deeper introduction.

2.4 Hyperparameters

As seen above, different families of kernel functions can depend on different types of *hyperparameters*. Selecting a suitable model for the prediction task at hand can be a complex task, and comprises of both selecting a kernel function that captures the dynamics of the data and finding hyperparameters for the adaption of the kernel of choice. For instance, for the SE kernel in (2.2) we want to find $\theta = (\sigma_f^2, \ell)^T$ making the kernel suitable for the given problem. One does often add a white noise parameter σ_n^2 to the kernel in applications. The parameter ℓ in K_{SE} is a "characteristic length-scale" [28], describing the distance to move along a dimension to make the outputs of a GP function uncorrelated. Furthermore, we can adopt an illustration made by Williams and Rasmussen by randomly draw two functions from a GP with the square exponential kernel. Here we use the 2 dimensional input and the more generalised

$$K_{SE}(x, x') = \sigma_f^2 \exp\left(-\frac{1}{2}(x - x')^T M (x - x')\right) + \sigma_n \delta, \quad (2.3)$$

with $\sigma_f^2 = 1$, $M = \text{diag}(\ell)^{-2}$ and zero noise term.

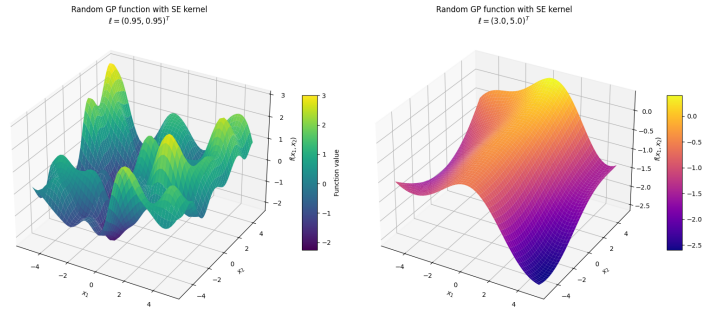


Figure 2.1: Two randomly drawn GP with $\ell = (0.95, 0.95)^T$ and $\ell = (3, 5)$, respectively.

In figure 2.1, ℓ 's role in K_{SE} for smoothness in GP functions becomes clear. This illustrates the endless variations GPs can be adapted to.

To conclude, GPs comprises a potential to capture a high variability of observed functions. The challenge of finding a suitable GP to a specific problem much depends on the choice of kernel and its hyperparameters. These can be set by evaluating the GP on a test set using cross validation. Furthermore, there are much to be said, and the reader is again referred to Rasmussen and Williams for a deeper and more comprehensive tutorial.

Chapter 3

Optimal Transport

Optimal Transport (OT) is a topic spanning many areas where some form of mass distribution is to be efficiently transported. This mass can consist of real world resources as in the STA case motivating this thesis, information in machine learning settings or probability distributions related to partial differential equations, to name a few. Naturally, it takes some energy to move these masses around which in an ideal setting is captured within a measurable cost function. Then one key follow-up question is – How can the transportation be made with a minimum cost? – which is a key question throughout this text as well as in Villani’s book [25], which is a major reference here.

This chapter begins by presenting necessary prerequisites in measure theory followed by a somewhat chronological introduction of OT; beginning with Monge’s Problem in 18th century France and continuing with today’s widely used formulation by Kantorovich. The introduction is then gradually transformed into the Unbalanced Optimal Transport (UOT) problem and the entropically regularised UOT which, informally, together with its duality formulation enables the masses that is to be moved and their corresponding targets to be of different weight as well as making the UOT problem convex, allowing for well-behaved solution algorithms such as the Sinkhorn algorithm introduced later on. Furthermore, a brief reformulation of the continuously presented problem statements to discrete versions is made to adapt for real world applications such as the case study made for the STA.

3.1 Measure theoretic preliminaries

In this section some prerequisite measure theory associated with OT is presented. In fact, OT deals with transporting measures, hence this is a natural starting point. The reader is assumed to have been exposed to some measure theory such as σ -algebras and probability measures.

For a metric space $(\mathcal{X}, d)$, we will denote the σ -ring generated by the class of all open sets of $\mathcal{X}$ as $\mathcal{B}(\mathcal{X})$ or short $\mathcal{B}$ which will be referred to as the *Borel σ -algebra*, and the set of σ -additive functions $\mu : \mathcal{B}(\mathcal{X}) \rightarrow \mathbb{R}$ as $\mathcal{M}(\mathcal{X})$ as in [1]. The introduction to OT will start by considering probability measures $\mathcal{P}(\mathcal{X}) = \{\mu \in \mathcal{M}(\mathcal{X}) : \mu(\mathcal{X}) = 1, \mu \geq 0\}$ and successively move to $\mathcal{M}(\mathcal{X})$ as the theory adopts more relaxed assumptions.

A discrete measure μ with weights μ_i and positions $x_i \in \mathcal{X}$ for $i = 1, \dots, n$ will be denoted as

$$\mu = \sum_{i=1}^n \mu_i \delta_{x_i}, \quad (3.1)$$

where δ_{x_i} is the Dirac delta function at x_i . Furthermore, μ is a discrete probability measure if $\sum_i \mu_i = 1$.

It is reasonable to begin by defining coupling since it is, as expressed by Villani [26] from where the definition is, a recurrent concept in OT.

Definition 3.1.1. (Coupling) Let $(\mathcal{X}, \mu)$ and $(\mathcal{Y}, \nu)$ be two probability spaces. Then (X, Y) is a coupling of (μ, ν) if X and Y are two random variables on some probability space $(\Omega, \mathbb{P})$ such that the law of X and Y is μ and ν , respectively.

Moreover, couplings can be interpreted as the construction of a single joint measure π on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν . i.e. if $\mu \in \mathcal{P}(\mathcal{X})$, $\nu \in \mathcal{P}(\mathcal{Y})$ and $\pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$, then for all measurable subsets $A \subset \mathcal{X}$, $B \subset \mathcal{Y}$ we have $\pi[A \times \mathcal{Y}] = \mu[A]$ and $\pi[\mathcal{X} \times B] = \nu[B]$. This is equivalent to saying that for every integrable and measurable functions φ, ψ on $\mathcal{X}, \mathcal{Y}$ the following holds

$$\int_{\mathcal{X} \times \mathcal{Y}} (\varphi(x) + \psi(y)) d\pi(x, y) = \int_{\mathcal{X}} \varphi d\mu + \int_{\mathcal{Y}} \psi d\nu.$$

Furthermore, we will denote $\Pi(\mu, \nu)$ as the set of all these measures π .

The notion of a push forward measure is a major part of OT; here defined in alignment with [1, 6].

Definition 3.1.2. A push forward measure $T_{\#} : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{M}(\mathcal{Y})$ is an operator defined as

$$T_{\#}\mu(B) := \mu(T^{-1}(B)), \quad \forall B \in \mathcal{B}(\mathcal{Y}),$$

where T is a measurable mapping.

The mapping $T^{-1}(B)$ is as usual the preimage $\{x \in \mathcal{X} : T(x) \in B\}$. Furthermore, T will be referred to as the *transport map*. Using this, we will sometimes for notational simplicity write $(p_{\mathcal{X}})_{\#}\pi = \mu$ and $(p_{\mathcal{Y}})_{\#}\pi = \nu$ instead of writing $\pi[A \times \mathcal{Y}] = \mu[A]$ and $\pi[\mathcal{X} \times B] = \nu[B]$, respectively, where $p_{\mathcal{X}} : (x, y) \mapsto x$, $p_{\mathcal{Y}} : (x, y) \mapsto y$ are the projection mappings.

For μ and ν as in definition 3.1.1 and $T(X) = Y$, one equivalent way of saying $T_{\#}\mu = \nu$ is that (X, Y) is a *deterministic coupling* [26].

Definition 3.1.3 (Deterministic coupling). We say that a coupling (X, Y) is deterministic if there exists a measurable function $T : \mathcal{X} \rightarrow \mathcal{Y}$ such that $Y = T(X)$.

Other measure theory related concepts of importance is the relative absolute continuity between measures, the theorem of Radon-Nikodym and the Lebesgue decomposition that for instance will be used when presenting divergences. The interested is referred to [7] for proofs of the theorems.

Definition 3.1.4 (Absolutely continuous w.r.t another measure). Let $(X, \mathcal{A}, \mu)$ be a measure space. A signed measure ν with domain $\mathcal{A}$ is said to be absolutely continuous with respect to μ , written $\nu \ll \mu$, if $\nu(E) = 0$ for every measurable set E where $|\mu|(E) = 0$.

Theorem 3.1.5 (Radon-Nikodym). *For a σ -finite measure $\nu \ll \mu$ on $\mathcal{A}$, where $(X, \mathcal{A}, \mu)$ is a σ -finite measure space, there exists a real-valued measurable function f on X such that*

$$\nu(E) = \int_E f d\mu \quad (3.2)$$

for every measurable set E where $\nu(E) < \infty$.

Definition 3.1.6 (Radon-Nikodym derivative). The function f in (3.2) is called the Radon-Nikodym derivative of ν with respect to μ and is written as

$$f = \frac{d\nu}{d\mu}, \quad \text{or} \quad d\nu = f d\mu. \quad (3.3)$$

Theorem 3.1.7 (Lebesgue decomposition). *For two σ -finite signed measures μ, ν in a measure space $(X, \mathcal{A})$, there exists σ -finite signed measures ν_0, ν_1 such that*

$$\nu = \nu_0 + \nu_1 \quad \nu_0 \perp \mu \quad \text{and} \quad \nu_1 \ll \mu.$$

This pair (ν_0, ν_1) is unique.

The notation $\mu \perp \nu$ indicates that μ and ν are mutually singular i.e. if there exists measurable sets $A, B \subset X$ such that $A \cup B = X$, $A \cap B = \emptyset$, $|\mu|(A) = 0$ and $|\nu|(B) = 0$. A proof of the Lebesgue decomposition can be found in [7].

3.2 Foundations of Optimal transport

Having established the measure theoretic preliminaries, this section focus on the basic foundations of the optimal transport problem. We begin by presenting the original problem stated by Monge and successively move to a relaxation of the problem made by Kantorovich, making OT more accessible for modern applications. The existence and uniqueness of solutions to the Kantorovich formulation is then introduced, followed by an informal presentation of the duality theory which, for instance, enables computational advantages and alternative economic interpretations.

3.2.1 Monge's Problem

Suppose a supplier has been assigned to move a certain volume of a resource from one place to another. Considering that the entire specified amount of resource is to be moved to another place one can further assume that the volume of the target space is equal to the volume to be moved, if there are no outer factors affecting these entities. This transportation of resource will likely have some associated cost in terms of money or the energy necessary for transportation, for instance. Thus, when all of the resources have been moved the transportation has caused some kind of total cost. Originally, these thoughts were presented by the French mathematician Monge in [13] who began the research of optimal transport, which aims to find the transportation plan minimising this total cost.

As mentioned, since the volumes of the domain and target are equal it is a convention to normalise these and consider probability measures $\mu \in \mathcal{P}(\mathcal{X})$

and $\nu \in \mathcal{P}(\mathcal{Y})$. Denoting the cost function as $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0} \cup +\infty$ Monge's problem then becomes to find a mapping T such that $T_{\#}\mu = \nu$ causing the lowest possible total cost. Formally, this minimisation problem can be expressed as follows [20]:

$$\inf \int c(x, T(x)) d\mu(x) : T_{\#}\mu = \nu, \quad (3.4)$$

hence the problem is to find a *deterministic optimal coupling*. A characteristic of Monge's problem is that it determines the destination y for each mass x , hence it does not consider sending a mass to more than one destination. Furthermore, one can see that there exists no solution to (3.4) if the problem is semi-discrete such that μ is discrete while ν is continuous.

3.2.2 Kantorovich's Optimal Transport

Nearly 150 years after Monge's problem was developed, a major contribution to OT was made by the Russian mathematician Kantorovich. Kantorovich introduced a so called relaxation of Monge's problem [25, 20] in his work [10], allowing for the mass to be split and sent to multiple destinations, and did actually get awarded with the Memorial Prize of Alfred Nobel in Economics for his work. The problem was then defined as the problem of finding the *optimal coupling*

$$\inf_{\pi \in \Pi(\mu, \nu)} \{I[\pi] := \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y)\}, \quad (3.5)$$

where $\Pi(\mu, \nu) = \{\pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) : (p_{\mathcal{X}})_{\#}\pi = \mu \text{ and } (p_{\mathcal{Y}})_{\#}\pi = \nu\}$.

Thus, Monge's formulations (3.4) can be seen as a special case of Kantorovich's problem (3.5) constraining the transport map T to take each mass x to a specific target y . Moreover, (3.5) can be interpreted in a probabilistic setting as well by noticing it is equivalent to

$$\inf \mathbb{E}[c(X, Y)], \quad (3.6)$$

for every pair of random variables $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ with μ and ν as their respective law.

Furthermore, Kantorovich's formulation makes the set of constraints convex, hence admits a duality theory [25]. To see the convexity of the set of couplings

we can simply take two couplings $\pi_1, \pi_2 \in \Pi(\mu, \nu)$, $\lambda \in [0, 1]$ and let $\pi = \lambda\pi_1 + (1 - \lambda)\pi_2$. Then for all measurable subsets $A \subset \mathcal{X}$, $B \subset \mathcal{Y}$ we have $\pi[A \times \mathcal{Y}] = \lambda\pi_1[A \times \mathcal{Y}] + (1 - \lambda)\pi_2[A \times \mathcal{Y}] = \lambda\mu[A] + (1 - \lambda)\mu[A] = \mu[A]$ and similar for ν and B , hence $\pi \in \Pi(\mu, \nu)$.

3.2.3 Duality of Kantorovich's formulation

The Kantorovich formulation (3.5) admits, as mentioned, a dual problem. This dual problem can be understood via the analogy "The shipper's problem" [25], which we will adapt here. Suppose that the transportation of mass costs $c(x, y)$ for each transportation from place x to y . Instead of making the transports yourself, a delivery company offers you they can handle the transportation for you by setting a pickup price $\varphi(x)$ and a delivery price $\psi(y)$ of the goods, and that the price $\varphi(x) + \psi(y)$ will always be less than the cost $c(x, y)$. They might even compensate you at some locations (accounting for negative prices). The essence of the duality is that the maximum potential profit the delivery company can make ($\sup \int \varphi d\mu + \int \psi d\nu$) is equal to the smallest possible transportation cost $I[\pi]$.

Letting $I(\pi)$ be as in Kantorovich's problem (3.5) and letting $J(\varphi, \psi) = \int_{\mathcal{X}} \varphi d\mu + \int_{\mathcal{Y}} \psi d\nu$ for functions $(\varphi, \psi) \in L^1(d\mu) \times L^1(d\nu)$ then Kantorovich's duality theorem states

$$\inf_{\Pi(\mu, \nu)} I[\pi] = \sup_{\varphi, \psi} J(\varphi, \psi), \quad (3.7)$$

where $\varphi(x) + \psi(y) \leq c(x, y)$. Hence, optimal transport can also be thought of as a way to determine optimal prices in order to maximising financial returns in a delivery plan. For a formal introduction of the duality theorem we refer the reader to any of [25, 26, 2].

3.3 Extensions and Regularisation

While the classical OT formulations are thorough and provide a robust foundation, there are often needs for extensions in practical settings. In this section we introduce so called φ -divergences which provides the OT toolbox with a method to quantify discrepancies and differences between measures. This naturally leads to the introduction of Unbalanced Optimal Transport (UOT) which is a further relaxation of OT allowing it to handle different

total masses. Finally, we finish this section by introducing the entropic regularisation which can be applied to both OT and UOT leveraging properties more suited for computational purposes.

3.3.1 φ -Divergences

φ -divergences are major tools used in extensions of classical OT. It is important to have a tool, particularly for unbalanced problems, penalising the transportation when the masses diverge from each other. This is what the φ -divergences used in OT is ought to do, and are therefore introduced here. The rest of this section will to a large extent follow [21, 15, 4] from which the following definitions are collected.

Definition 3.3.1. An entropy function is a function $\varphi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R} \cup \{\infty\}$ that is convex, lower semi continuous and satisfies $\text{dom } \varphi \cap \mathbb{R}_{>0} \neq \emptyset$. They further define the speed of growth at ∞ as

$$\varphi'_\infty = \lim_{x \rightarrow +\infty} \varphi(x)/x \in \mathbb{R} \cup \{\infty\}.$$

An entropy function φ is said to be super linear if $\varphi'_\infty = \infty$, meaning it grows faster than any linear function. The Lebesgue decomposition, stating that for a given measure ν , (a σ -finite measure) μ has a unique decomposition $\mu = \mu^s + \mu^\perp$ where $\mu^s \ll \nu$, is used in the following definition.

Definition 3.3.2. Let $\frac{d\mu}{d\nu}\nu + \mu^\perp$ be the Lebesgue decomposition of μ with respect to ν . The divergence with respect to an entropy function φ is

$$\mathcal{D}_\varphi(\mu \mid \nu) := \begin{cases} \int_{\mathcal{X}} \varphi\left(\frac{d\mu}{d\nu}\right) d\nu + \varphi'_\infty \mu^\perp(\mathcal{X}), & \text{if } \mu, \nu \geq 0, \\ \infty, & \text{otherwise.} \end{cases} \quad (3.8)$$

Divergences are useful in a broad spectrum of data science where one wants to compare measures such as distributions of data [21]. For instance, a divergence can be used as a loss function as an alternative to usual metric distances; they provide a notion of how close two measures are but do not necessarily admit the triangle inequality nor the symmetry property. The Kullback-Leibler divergence is particularly popular in UOT and is defined as the divergence of the Shannon-entropy.

Definition 3.3.3 (Kullback-Leibler divergence).

$$\mathcal{D}_{KL}(\mu \mid \nu) := \begin{cases} \int_{\mathcal{X}} \log\left(\frac{d\mu}{d\nu}(x)\right) d\mu(x) - \int_{\mathcal{X}} d\nu + \int_{\mathcal{X}} d\mu & \text{if } \mu^\perp = 0, \\ +\infty & \text{otherwise.} \end{cases} \quad (3.9)$$

Here the entropy function is the Shannon-Boltzman entropy

$$\varphi_{KL}(x) = \begin{cases} x \log x - x + 1 & x > 0, \\ 1 & x = 0, \\ +\infty & \text{otherwise,} \end{cases} \quad (3.10)$$

and $\varphi'_\infty = \lim_{x \rightarrow +\infty} (x \log(x) - x + 1)/x = +\infty$, which then is said to be a superlinear entropy function. The case when $\mu^\perp = 0$ is when $\mu \ll \nu$ which can be seen since $\mu \ll \nu$ only if $\mu^\perp \ll \nu$, and if $\mu \ll \nu$ and $\mu \perp \nu$, then $\mu^\perp = 0$.

The convex conjugate or Legendre transform $\varphi^* : \mathbb{R} \rightarrow \mathbb{R}$ of an entropy function φ is written as $\varphi^*(p) = \sup_{q \geq 0} \{pq - \varphi(q)\}$ and will be used in the Dual formulation of UOT as well as in the Sinkhorn algorithm. This conjugate has the following useful properties [22]:

1. φ^* is non-decreasing,
2. $\text{dom}(\varphi^*) = (-\infty, \varphi'_\infty)$,
3. $\lim_{q \rightarrow -\infty} \varphi^*(q) = -\varphi(0)$ and $\lim_{q \rightarrow +\infty} \varphi^*(q) = +\infty$.

Furthermore, one can multiply a divergence with a parameter $\rho > 0$ which in UOT corresponds to the "characteristic radius of transportation" [21]. One can then make use of the fact that $\rho \mathcal{D}_\varphi = \mathcal{D}_{\varphi\rho}$. Moreover, it can be verified that the p solving $\varphi^*(q) = \sup_p pq - \varphi(p)$ for the Kullback-Leibler entropy function $\rho \varphi_{KL}(p) = \rho(p \log p - p + 1)$ is $p = e^{q/\rho}$ and thus the Legendre transform of $\varphi_{KL}(p)$ is $\varphi_{KL}^*(q) = \rho(e^{q/\rho} - 1)$.

3.3.2 Unbalanced Optimal Transport

Due to the restriction of OT the fact that the measures must have the same total mass $\int d\mu = \int d\nu$ or else the set of couplings $\Pi(\mu, \nu) = \emptyset$ is empty, there are countless scenarios where one cannot use OT in. Simply any case where one wants to push some data or mass from a measure of different mass than the target. Furthermore, when a measure contains outliers, then

classical OT forces these outliers to be transported, hence OT is sensitive to outliers in the distribution to be pushed.

UOT tackles these issues and restrictions by incorporating φ -divergences in a relaxation of Kantorovich's problem (3.5) and letting the measures not being constrained to probability spaces but rather $\mu \in \mathcal{M}^+(\mathcal{X})$ and $\nu \in \mathcal{M}^+(\mathcal{Y})$, and allowing for mass points to be "destroyed" and "created" rather than moved. There are several types of UOT such as static, dynamic and conic [21]. Here the focus stays within the static framework, however, the application to the STA case study would likely not suffer from using the dynamic setting, hence, that might be of interest in the future. Furthermore, [5] shows that the dynamical setting actually is equivalent to static UOT formulated with semi-couplings.

This section will mainly follow [21, 15, 5]. The definition of the UOT problem follows.

Definition 3.3.4. The Unbalanced Kantorovich problem is defined as the minimisation problem over transport plans $\pi \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y})$ of

$$\inf_{\pi \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y})} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \mathcal{D}_{\varphi_1}((p_{\mathcal{X}})_{\#}\pi \mid \mu) + \mathcal{D}_{\varphi_2}((p_{\mathcal{Y}})_{\#}\pi \mid \nu). \quad (3.11)$$

Note that it is no longer the case that $(p_{\mathcal{X}})_{\#}\pi = \mu$ nor $(p_{\mathcal{Y}})_{\#}\pi = \nu$ and $\mathcal{D}_{\varphi_1}((p_{\mathcal{X}})_{\#}\pi \mid \mu)$ provides a notion of how much $(p_{\mathcal{X}})_{\#}\pi$ deviates from μ . As mentioned in [21], while $(p_{\mathcal{X}})_{\#}\pi$ corresponds to the transported mass, the ratio $(p_{\mathcal{X}})_{\#}\pi \mid \mu$ in the divergence can be interpreted as the mass destruction (or creation).

3.3.3 Entropic regularisation

Finding an exact solution to a OT problem is not always straight forward and can be computationally expensive for large-scale problems. Furthermore, the optimal transport plan might not be unique in UOT, hence there is a need for approximate solutions. By adding an entropic regularisation penalty, one ensures the problem is convex and allowing for highly efficient solution algorithms such as the Sinkhorn algorithm, hence both the uniqueness and the computational issues are addressed.

We begin by introducing the regularisation term influenced by [21, 15].

Definition 3.3.5. For two positive measures $\mu, \nu \in \mathcal{M}(\mathcal{X})$, the entropic regularisation of the UOT problem is defined as

$$\inf_{\pi \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y})} \mathcal{UOT}_\varepsilon, \text{ where}$$

$$\mathcal{UOT}_\varepsilon := \int_{\mathcal{X} \times \mathcal{Y}} c d\pi + \mathcal{D}_{\varphi_1}((p\mathcal{X})_\# \pi \mid \mu) + \mathcal{D}_{\varphi_2}((p\mathcal{Y})_\# \pi \mid \nu) + \varepsilon \mathcal{D}_{KL}(\pi \mid \mu \otimes \nu), \quad (3.12)$$

where $\mu \otimes \nu$ denotes the product measure such that for $A \in \mathcal{X}$ and $B \in \mathcal{Y}$ we have $\mu \otimes \nu[A \times B] = \mu[A]\nu[B]$.

As mentioned previously, when using the Sinkhorn algorithm the duality of OT is used. In particular, the duality of the entropic regularisation of the unbalanced OT is used. This dual formulation is based on [21] which in turn is based on the Fenchel-Rockafellar duality theorem in [18]. In comparison with the Kantorovich duality (3.7), we here optimising over the continuous functions $(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})$ also called the *dual potentials* [22].

The duality of the entropic UOT reads

$$\mathcal{D}_\varepsilon(f, g) := - \int_{\mathcal{X}} \varphi_1^*(-f(x)) d\mu(x) - \int_{\mathcal{Y}} \varphi_2^*(-g(y)) d\nu(y) - \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} (e^{(f(x)+g(y)-c(x,y))/\varepsilon} - 1) d\mu(x) d\nu(y), \quad (3.13)$$

where $\varphi_i^*(p) = \sup_{q \geq 0} \{pq - \varphi(p)\}$ is the convex conjugate or Legendre transformation. For notational purpose we follow [22] and continuing with $\mu \otimes \nu$ for $\mu[x]\nu[y]$ as above and write $\langle \mu, f \rangle_{\mathcal{X}}$ for $\int_{\mathcal{X}} f d\mu(x)$, $f \oplus g$ for $f(x) + g(y)$ and get the dual problem as

$$\sup_{(f,g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \mathcal{D}_\varepsilon(f, g)$$

$$\mathcal{D}_\varepsilon(f, g) := -\langle \mu, \varphi_1^*(-f) \rangle_{\mathcal{X}} - \langle \nu, \varphi_2^*(-g) \rangle_{\mathcal{Y}} - \varepsilon \langle \mu \otimes \nu, e^{\frac{f \oplus g - c}{\varepsilon}} - 1 \rangle. \quad (3.14)$$

The equality of the dual (3.14) and the entropic UOT (3.12) holds for a superlinear entropy function φ [11], hence for the Kullback-Leibler divergence.

When the optimal pair (f^*, g^*) has been found, then the optimal transportation plan is found by simply computing

$$\pi^*(x, y) = e^{\frac{1}{\varepsilon}(f^*(x)+g^*(y)-c(x,y))} \mu(x)\nu(y) \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y}). \quad (3.15)$$

3.4 Computational Optimal Transport

As the theoretical foundations and the relevant extensions have been established, it is time for an introduction to the practical and computational aspects of OT. Since data is naturally discrete in typical real world scenarios, it is required to adapt the continuous theory to a discrete setting. Thus, this section introduce the discrete formulation of OT, in a similar order as in the continuous case, subsequently followed by an introduction of the Sinkhorn algorithm. The Sinkhorn algorithm is an efficient iterative method specifically created for the discrete entropic OT problem. Here, the Sinkhorn algorithm will be designed via the discrete entropic UOT.

3.4.1 Discrete Optimal Transport

In the discrete setting of OT, the measures μ and ν can be interpreted as in (3.1) as histograms – i.e., weighted sums – $\mu = \sum_{i=1}^n \mu_i \delta_{x_i}$ and $\nu = \sum_{j=1}^m \nu_j \delta_{y_j}$, where δ_{x_i} is the delta function and $\mu_i \in \mathbb{R}_{\geq 0}, \nu_i \in \mathbb{R}_{\geq 0}$. Furthermore, these measures could just as well be vector valued measures, then they would, for instance, take the form $\mu_i = (\mu_i^{r_1}, \dots, \mu_i^{r_k})^T \in \mathbb{R}_{\geq 0}^k$, and similar for ν_j , for resources $r_1, \dots, r_k$ in the STA case study. This section details the discrete formulations of Monge (3.4), Kantorovich (3.5), UOT, entropic UOT (3.12) and their duals.

Monge's problem (3.4) becomes to find a mapping $T : \{x_1, \dots, x_n\} \rightarrow \{y_1, \dots, y_m\}$ such that $T_{\#}\mu = \nu$ which every mass transported to a point y_j must equal the mass ν_j :

$$\nu_j = \sum_{i:T(x_i)=y_j} \mu_i, \quad \forall j \in \{1, \dots, m\};$$

in other words, the problem of Monge is

$$\inf_T \left\{ \sum_i^n c(x_i, T(x_i)) : T_{\#}\mu = \nu \right\}. \quad (3.16)$$

Furthermore, in the particular case when the weights are uniform and $n = m$ Monge's problem becomes the *optimal assignment* problem where one searches for a bijection σ in the symmetric group $\mathbf{S}_n$ solving the following minimisation [15]:

$$\inf_{\sigma \in \mathbf{S}_n} \left\{ \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)} : \sum_{i \in \sigma^{-1}(j)} \mu_i = \nu_j \right\}. \quad (3.17)$$

As indicated previously, Monge's problem does not have a solution if $m > n$.

The discrete interpretation of Kantorovich's problem (3.5) becomes

$$\inf_{\pi \in \Pi(\mu, \nu)} \sum_i^n \sum_j^m C_{ij} \pi_{ij}, \quad (3.18)$$

where in this case $\Pi(\mu, \nu) = \{\pi \in \mathbb{R}_{\geq 0}^{n \times m}, \sum_j \pi_{ij} = \mu_i, \sum_i \pi_{ij} = \nu_j\}$.

Continuing with the same measures as above, the entropic UOT in (3.12) becomes in the discrete case

$$\inf_{\pi \in \mathbb{R}_+^{m \times n}} \left\{ \sum_{i=1}^n \sum_{j=1}^m C_{ij} \pi_{ij} + \mathcal{D}_{\varphi_1}(\pi \mathbf{1}_m \mid \mu) + \mathcal{D}_{\varphi_2}(\pi^T \mathbf{1}_n \mid \nu) + \varepsilon \mathcal{D}_{KL}(\pi \mid \mu \otimes \nu) \right\}, \quad (3.19)$$

where $\mathbf{1}_n$ is a vector of ones.

Finally, in the dual case we also consider the dual potentials as vectors $f \in \mathbb{R}^n, g \in \mathbb{R}^m$ where $f_i = f(x_i)$ and $g_j = g(y_j)$. Moreover, recall the direct formula of the optimal plan (3.15) which in the discrete case becomes $\pi_{ij} = \exp\left\{\left(\frac{1}{\varepsilon}(f_i + g_j - c_{ij})\right)\right\} \mu_i \nu_j$. Hence the dual formulation of (3.14) is

$$\begin{aligned} & \sup_{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m} \mathcal{D}_\varepsilon(f, g) \\ \mathcal{D}_\varepsilon(f, g) &:= - \sum_{i=1}^n \varphi_1^*(-f_i) \mu_i - \sum_{j=1}^m \varphi_2^*(-g_j) \nu_j - \varepsilon \sum_{ij} \left(e^{\frac{f_i + g_j - c_{ij}}{\varepsilon}} - 1 \right) \mu_i \nu_j. \end{aligned} \quad (3.20)$$

3.4.2 The Sinkhorn Algorithm

Having established the discrete optimal transport it is natural to introduce the computational aspects of the thesis. To find the optimal dual potentials for the dual problem (3.14) we will implement the Sinkhorn algorithm. A central part of the Sinkhorn algorithm is the *Softmin operator* which is used to define two maps directly used in the algorithm [22, 21].

Definition 3.4.1. The Softmin operator $Smin_\mu^\varepsilon : \mathcal{C}(\mathcal{X}) \rightarrow \mathbb{R}$ is for any continuous function $f \in \mathcal{C}(\mathcal{X})$ defined as

$$Smin_\mu^\varepsilon(f) := -\varepsilon \log \langle \mu, e^{(-f/\varepsilon)} \rangle, \quad (3.21)$$

for any $\mu \in \mathcal{M}^+(\mathcal{X})$, $\varepsilon > 0$.

Furthermore, in the discrete setting the Softmin operator is formulated as

$$Smin_\mu^\varepsilon(f) = -\varepsilon \log \sum_i^n e^{(-f_i/\varepsilon + \log \mu_i)}. \quad (3.22)$$

In computations one can make use of the Log-Sum-Exponential function

$$LSE_i^n := u_* + \log \sum_i^n e^{(u_i - u_*)} = \log \left(e^{u_*} \sum_i^n e^{(u_i - u_*)} \right) = \log \sum_i^n e^{u_i} \quad (3.23)$$

when implementing the Softmin operator, where $u_* = \max_i u_i$ of the vector u . Since $u_i - u_* \leq 0$, we have that $e^{u_i - u_*} \leq 1$ (and $LSE_i^n(u) \leq u_* + \log n$), which prevents numerical explosions in the exponential [21]. This will be used in the implementation of the Sinkhorn algorithm.

The two mappings incorporating the Softmin operator are defined as follows

$$\begin{aligned} \mathcal{S}_\mu(f)(y) &:= Smin_\mu^\varepsilon(c(\cdot, y) - f), \\ \mathcal{S}_\nu(g)(x) &:= Smin_\nu^\varepsilon(c(x, \cdot) - g), \end{aligned} \quad (3.24)$$

taking $\mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) \rightarrow \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})$. These operators are the main and central parts of both OT and UOT updates in the Sinkhorn algorithm, however, an additional anisotropic operator in the UOT case is necessary.

Definition 3.4.2. The anisotropic proximity operator is defined for any $p \in \mathbb{R}$ as

$$Approx_h^\varepsilon(p) := \arg \min_{q \in \mathbb{R}} \varepsilon e^{\frac{p-q}{\varepsilon}} + h(q) \in \text{dom}(h) \cup \{+\infty\}, \quad (3.25)$$

where $h : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function and $\varepsilon > 0$.

This anisotropic operator is applied to $\mathcal{S}_\mu(f)(y)$ which actually are the optimality conditions for the entropic dual (3.14), and will thus be used for updates of f and g in the Sinkhorn algorithm. The first order optimality conditions are applied pointwise and are formulated as

$$\begin{aligned} f(x) &= -\text{Aprox}_{\varphi^*}^\varepsilon(-\mathcal{S}_\nu(g)(x)), \quad \mu - a.e., \\ f(x) &= -\text{Aprox}_{\varphi^*}^\varepsilon(-\mathcal{S}_\mu(f)(y)), \quad \nu - a.e., \end{aligned} \tag{3.26}$$

where $\mathcal{S}_\mu(f)(y)$ is as in (3.24).

As mentioned, these are used in the Sinkhorn algorithm for the updates of the dual potentials f, g .

Definition 3.4.3. The Sinkhorn algorithm is defined as updating the dual potentials as follows

$$\begin{aligned} f_{t+1} &:= \arg \max_f \mathcal{D}_\varepsilon(f, g_t) = -\text{Aprox}_{\varphi^*}^\varepsilon(-\mathcal{S}_\nu(g_t)(x)), \\ g_{t+1} &:= \arg \max_g \mathcal{D}_\varepsilon(f_{t+1}, g) = -\text{Aprox}_{\varphi^*}^\varepsilon(-\mathcal{S}_\mu(f_{t+1})(y)), \end{aligned} \tag{3.27}$$

starting at some $g_t \in \mathcal{C}(\mathcal{X})$.

Finally, one important fact proved in [22] is that a pair of dual potentials (f, g) is optimal if and only if it is a fixed point of the Sinkhorn mapping (3.27). The fact that they are fixed point mappings is based on the non-expansiveness and non-decreasing behaviour of the $\text{Aprox}_{\varphi^*}^\varepsilon$ -operator (i.e., they are contractions), also referred to [22] for a proof.

We have now established the essentials for presenting the full Sinkhorn algorithm, which is presented here.

Algorithm 1 The Sinkhorn Algorithm

Input: Cost matrix $\{C_{ij}\}_{i,j=1}^{n,m} = \{c(x_i, y_j)\}_{i,j=1}^{n,m} \in \mathbb{R}^{n \times m}$, source $\mu = \sum_{i=1}^N \mu_i \delta_{x_i}$ and target $\nu = \sum_{j=1}^M \nu_j \delta_{y_j}$, where $\mu_i \in \mathbb{R}^n, \nu_j \in \mathbb{R}^m$ and vectors $x_i \in \mathbb{R}^D, y_j \in \mathbb{R}^D$ for $i = 1, \dots, n, j = 1 \dots, m$.

Parameters: Entropy φ and regularisation $\varepsilon > 0$.

Output: Vectors $f \in \mathbb{R}^n$ and $g \in \mathbb{R}^m$ approximating the dual potentials of (3.14).

```
function SINKHORN( $C, \alpha, \beta$ )  
   $f \leftarrow 0_n, g \leftarrow 0_m$   
  while updates > tolerance do  
     $g_j \leftarrow -\varepsilon LSE_{i=1}^n [\log \mu_i + (f_i - C_{ij})/\varepsilon], \forall j \in \{1, \dots, m\}$   
     $g_j \leftarrow -\text{Approx}_{\varphi^*}^{\varepsilon}(-g_j), \forall j \in \{1, \dots, m\}$   
     $f_i \leftarrow -\varepsilon LSE_{j=1}^m [\log \nu_j + (g_j - C_{ij})/\varepsilon], \forall i \in \{1, \dots, n\}$   
     $f_i \leftarrow -\text{Approx}_{\varphi^*}^{\varepsilon}(-f_i), \forall i \in \{1, \dots, n\}$   
  end while  
  return  $f, g$   
end function
```

One should mention that the Sinkhorn algorithm can be written in matrix-vector products allowing for faster computations, especially since it then can be implemented on GPUs [4].

3.5 Applications of Optimal Transport

Apart from theoretical use in mathematics, OT, its extensions and the Wasserstein spaces and distances that will be introduced here have various applications in areas such as gradient flows, image processing, machine learning, colour transferring and signal processing, to name a few. To demonstrate this, we will now introduce the Wasserstein metric (also called the Earth Mover Distance) and subsequently present some of their applications.

3.5.1 Wasserstein distances

The total optimal transportation cost $C(\mu, \nu) = \inf I[\pi]$ can be thought of as a type of distance in some settings, although, not strictly speaking a metric. There are, however, ways to make this cost a metric in the general sense.

One of the most established is the Wasserstein distance W_p , which can be a quite simple construction by defining the cost function as $c(x, y) := d(x, y)^p$, where d is a metric on $\mathcal{X}$ [26]. As in [26], we define the Wasserstein distance as follows.

Definition 3.5.1. Let $(\mathcal{X}, d)$ be a Polish (complete and separable) metric space and $p \in [1, \infty)$. The Wasserstein distance of order p is for any probability measures $\mu, \nu \in \mathcal{P}(\mathcal{X})$ defined as

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p} \quad (3.28)$$

$$= \inf \{ [\mathbb{E} d(X, Y)^p]^{1/p}, \text{ law}(X) = \mu, \text{ law}(Y) = \nu \}. \quad (3.29)$$

Furthermore, this is not strictly a distance unless one restrain it such that it takes on finite values only. One way of creating sufficient restriction is by defining the Wasserstein space which equipped with the Wasserstein distance becomes a metric space.

Definition 3.5.2. With $(\mathcal{X}, d)$ and p as in 3.5.1, the Wasserstein space of order p is defined as

$$P_p(\mathcal{X}) := \{ \mu \in \mathcal{P}(\mathcal{X}) : \int_{\mathcal{X}} d(x_0, x)^p \mu(dx) < +\infty \}, \quad (3.30)$$

where $x_0 \in \mathcal{X}$ is arbitrary.

Hence we now have a distance that in words measures the minimal energy needed to move the probability mass from one measure or distribution to another; at least when $p = 1$ which is also called the Kantorovich-Rubinstein distance.

3.5.2 Applications of Optimal Transport and the Wasserstein Distance

Wasserstein and OT distances can successfully be implemented as objective loss functions in machine learning as well as in complex inverse problems. For instance, [8] use regularisations of the Wasserstein distance as loss function in a multi-label classification setting leveraging the abilities of the Wasserstein

penalty to promote smoother predictions. Furthermore, one can leverage the Wasserstein metric in a similar approach but in an unsupervised setting where the aim is to learn dictionary elements by applying the Wasserstein metric to compare feature similarities instead [19].

The learning of seismological activity can leverage implementing the Wasserstein metric [30]. In particular, the Wasserstein-Fisher-Rao (WFR) metric can be applied to the earthquake localisation problem to improve upon the already successful use of the quadratic Wasserstein metric, making the method more robust to intense noise. The WFR was developed in a UOT setting and combines the Wasserstein distance with the Fisher-Rao metric allowing for so called teleportation (destruction and creation) of mass as well as the transportation [3].

Another application regarding gradient flows is the modelling of pedestrian flow and dynamics of crowds during, for instance, emergency evacuations in a macroscopic perspective [12]. There, they model the evolution of a crowd’s density as a gradient flow in a Wasserstein space. In particular, the crowds are modelled with a time varying density $\rho(x, t)$ instead of as in a microscopic view treating each individual as discs.

Finally, fitting Gaussian Mixture Models (GMMs) can be computationally challenging. In [29] it is demonstrated that incorporating the WFR geometry can increase efficiency of algorithms for this task, operating directly on the space of probability measures. In particular, they propose an algorithm computing the Non-Parametric Maximum Likelihood Estimate (NPMLE) in a Gaussian mixture model leveraging both convergence properties and computational gain implementing a gradient descent defined via the WFR over the Wasserstein space, often realised using particle systems updating particle locations as well as weights.

Chapter 4

A Simulated Resource Allocation Problem for the Swedish Transport Administration

4.1 Problem Recap

Recall the challenges outlined in the introduction. Large construction projects involve a significant amount of resources and material transport, which might include insufficient reuse of excavated resources and, hence, has an insufficient impact on sustainability. In this case study, mathematical methods are implemented to explore their potential applicability addressing this issue. In particular, Gaussian Processes and Optimal Transport is applied, providing a simulated scenario that attempts to reflect the STA's need for resource management. The STA's database ELSA aims to collect comprehensive activity data from projects which can facilitate predictions and forecasts as well as statistical analyses in the future, thus also GP and OT. While ELSA is not directly used in this thesis, the simulated case study presented here aims to explore some of the methodologies that can be relevant for future usage of the ELSA database.

GPs is used because of the natural fitting in this setting, due to not nec-

essarily purely time dependent data but rather seasonalities and trends can be captured in some feature vectors instead. Although, it is possible there are methods more prominent in capturing the variations of the data such as time dependent state space GPs that might be able to capture dependencies between states in resource demand and supply over time, if that exists. Here we, however, assume independency between states since it seems natural that the amount of resources excavated in one location does not affect the resources excavated in another construction site. In particular, this is likely the case since no resources are currently known, to us, to be recycled in between projects. Furthermore, neural networks or gradient boosting machines might also be able to capture the complexities of the data, however, GP is chosen much because of the explainable models. With this said, it might be of interest to switch to one of the others in the future when data has grown and if the results of this study shows to be useful.

The GP predictions of resource amounts will be based on the generated data detailed in appendix and described in section 2.1. Moreover, the predicted means of resource demand and supply on each construction site will be used as input to the OT implementation. This can likely be extended such that the entire prediction interval is used as input to a *Robust* OT (or UOT) implementation that outputs worst and best case scenarios. Here, entropic regularised UOT will be used, considering the measures of the supply and demand are naturally different, and solved using the Sinkhorn algorithm (1). One prominent challenge is the construction of a cost function. To capture sustainable impact is a hard task but one can argue that the shorter distance travelled the less negative impact it has, although, plausibly not entirely true. The Haversine distance can therefore probably reflect much of the cost and will be used here, however, a distance recovered from Google maps' API would most likely improve this significantly.

The methodology of this case study is as follows. Given a major project, or projects, consisting of several construction sites, the goal is to predict the resource surplus and deficit at each construction site over time. The total of the surplus predictions are then compared to prior estimates made by the constructor and the STA to evaluate a possible incorporation of mathematical methods in the STA's prediction toolbox. Furthermore, an optimal transport plan of the predicted resources at a chosen time interval is computed via the Sinkhorn algorithm. With that said, this plan can possibly be enhanced to incorporate a more dynamic setting in OT in the future. The simulation

can model a given major project or perhaps a subset of construction sites included in one or more main projects where each have their own timeline.

4.2 STA Dataset: Exploration and Initial Analysis

We begin by exploring the data. In particular, features of interest such as the distribution of projects and their associated construction sites and the contractor's and the STA's prior estimates of excavated resources. The estimates are plotted against the actual outcome of resources in tonnes. Also, a scatterplot of the errors from the estimates are plotted, together with a LOWESS fitted regressor to provide insight of the estimates, as well as error distribution plots.

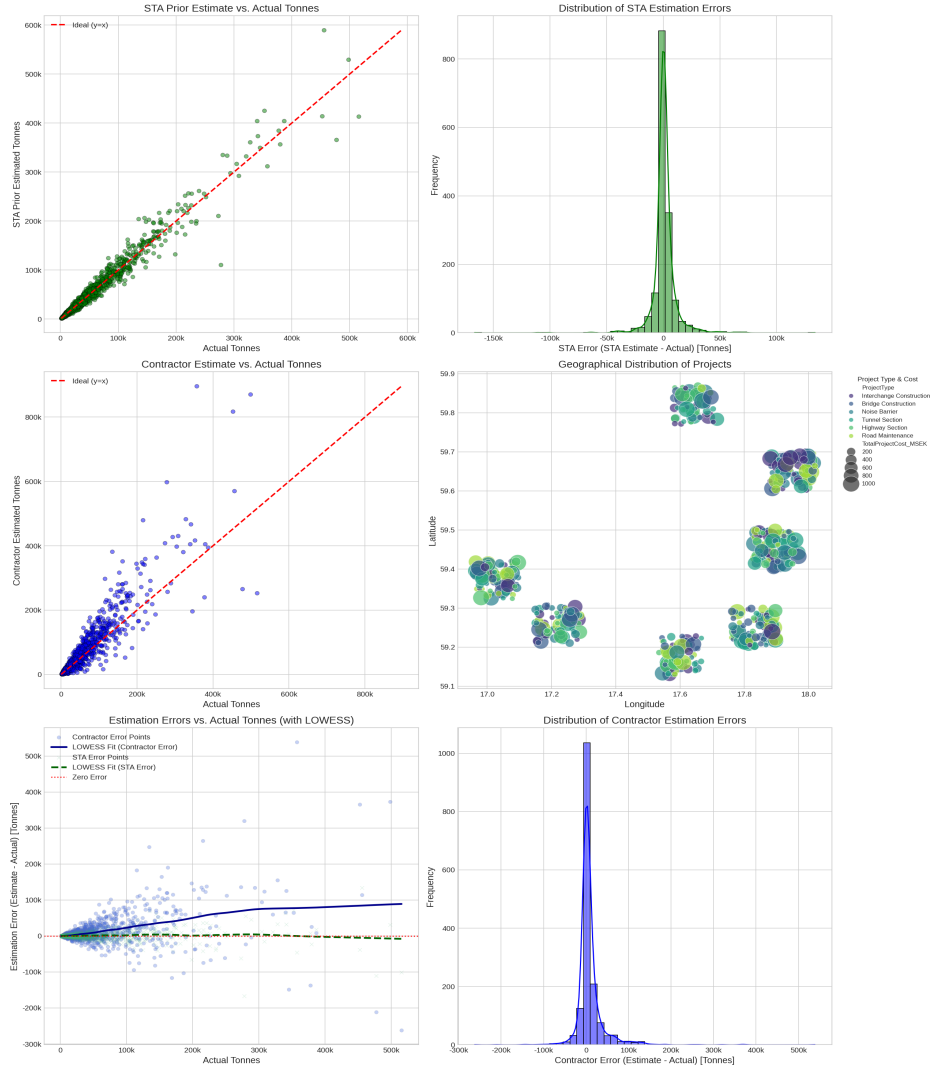


Figure 4.1: From left to right and top to bottom: STA's estimated tonnes vs actual tonnes with line of equality; distribution of STA errors; Contractor's estimated tonnes vs actual tonnes with line of equality; locations of construction sites for each main project in latitude and longitude with size of point representing size of construction; scatter plot of contractor's and STA's estimation errors with a fitted LOWESS regressor; distribution of contractor's errors.

One trend to be spotted above is that the contractor's relative error seems to be greater in average than the STA's, which by eye looks quite accurate, although the error distribution shows their estimates can differ as much as 50.000 tonnes and 100.000 for the contractor's estimates. There are clear heteroscedastic behaviour of the contractor's estimates in relation to the actual tonnes, which one might had expected. This is apparent for the STA's estimates as well, although not as prominent.

Furthermore, the response variables, the excavations and resource demand are plotted over time; including resources needed to be transported to depot i.e. those resources that cannot be recycled due to toxicity. This illustrates the need for properly managed bulk materials. The plots are created for the entire lifespan of all projects involved, hence the distribution. In addition, the cumulative development of the total amount of resources, illustrating how fast they add up if not continuously reallocated or reused is plotted.

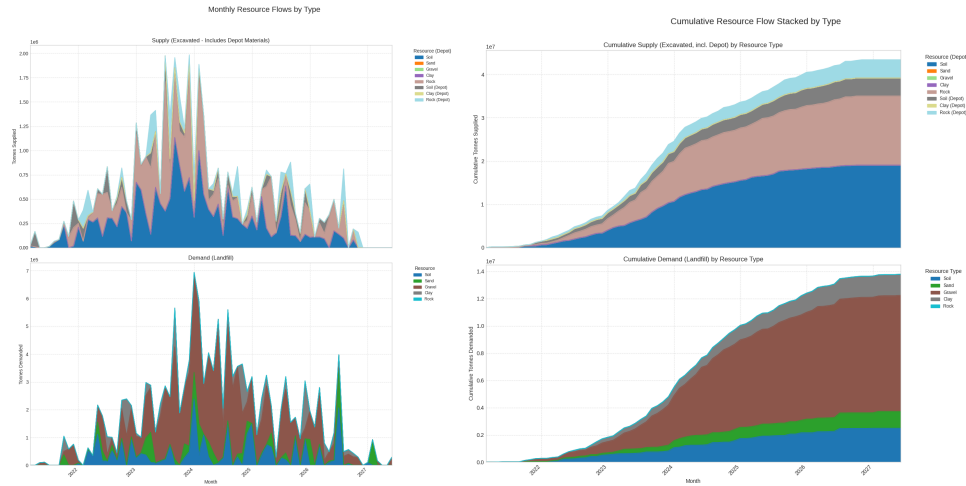


Figure 4.2: Time series plots of excavated and landfill used resources and the cumulative surplus and demand of resources.

As seen, there are vast amounts of resources needed to be handled. Hence, preparation is crucial in order to manage this properly. Suppose, for instance, that rock can be crushed into finer types as discussed earlier, allowing for reuse in multiple categories, not only as landfill but in pavement foundation works and possibly as aggregate in used concrete. Furthermore, after discussion with people directly involved with such bulk material management,

it has come to light that without prior knowledge of when certain resource types are being excavated the STA might have to pay for these resources to be removed rather than selling refined rock to constructors and entrepreneurs with a demand. Thus, it is crucial to have an as accurate as possible understanding of when rock and other resources might be excavated and, thence, so is forecasting future resource appearance.

4.3 GPR Prediction and Forecast

Having established the importance of accurate predictions, it is now time for developing a model built for making forecasts of resource appearance. The predictive model is trained, evaluated and implemented using the GPflow package using TensorFlow in Python. The the data set was chronologically cut into the training and test set containing the first 75% and the last 25% of the data, respectively. To account for the heteroscedasticity in the data a logarithmic transformation was made, since one assumption in the GP model used is non-heteroscedasticity. However, one possible improvement might be if one uses variational inference (VI) within the training, which also is making use of the Kullback-Leibler divergence (3.9), hence accounting for the larger variations. However, the log transform does capture this variance behaviour quite effectively as seen in the evaluation plot 4.3. Furthermore, using a Matérn52 kernel and making a hyperparameter optimisation grid search over 500 parameters, the trained model's forecast improves the prior estimates made by the contractor as well as the STA on the test set.

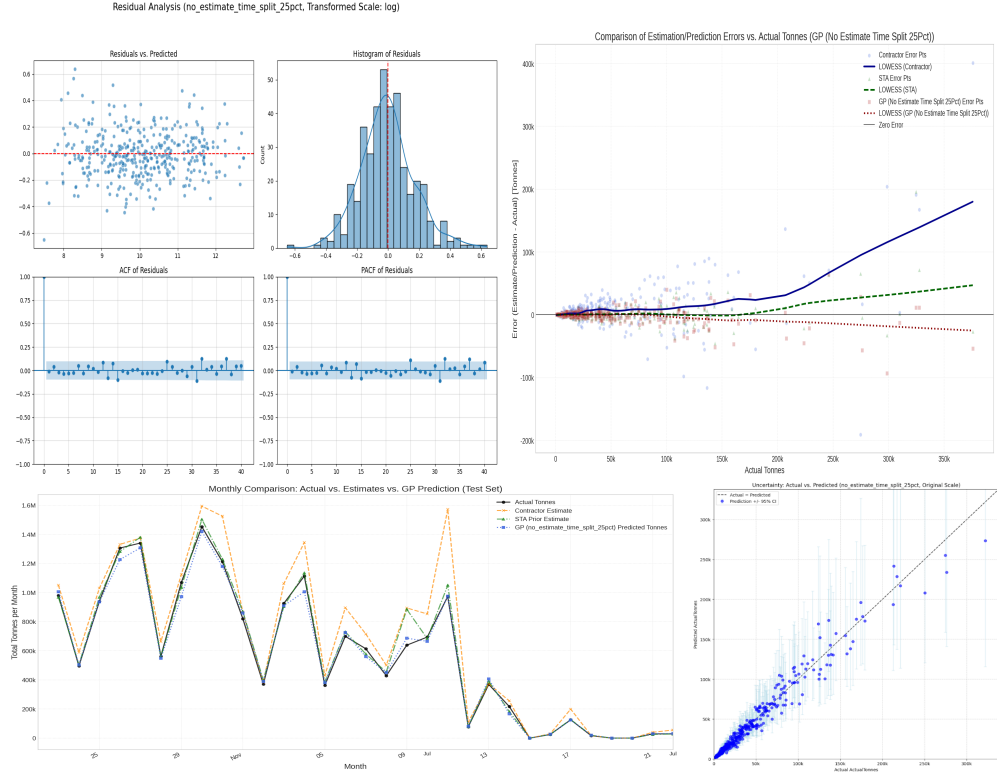


Figure 4.3: From right to left and top to bottom: Error analysis of the log transformed data showing error distribution as well as ACF and PACF; The prediction error vs the prior estimates' errors; forecast together with the prior estimates on test set; forecast Vs. actual values with 95% CI.

The ACF and PACF show no signs of linear correlation between states and the error distribution looks normal up to an additive constant, hence it can be assumed most of the true function's behaviour has been captured by the GP. The forecast shown in 4.3 also suggest a quite good fit. Comparing it with the prior estimates made by the constructor and the STA one can detect slightly more accurate predictions on large excavations than the STA's estimates, while the STA seems to outperform this model on smaller excavations. The contractor is here outperformed entirely by the GP forecast. However, there are a significant uncertainty in the forecast, which is illustrated by the 95% confidence interval in 4.3. This could possibly be narrowed further by feature engineering further or incorporating VI as mentioned. In conclusion,

considering that this is a prototype predictive model, forecasting resources in large construction sites using statistical and machine learning models has the potential to bring extra nuance and foundation to the STA’s decision making. Particularly, if further and deeper analysis is being made together with an exploration and evaluation of further predictive models that might be more tailored to this type of forecast.

4.4 Optimal Resource Allocation using Optimal Transport

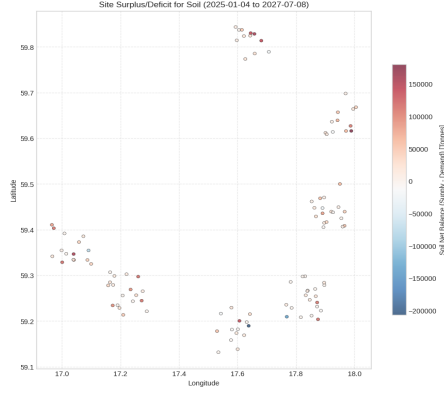
Applying the GPR model to a given set of future projects containing the same information as the trained data, we can make a forecast of demand and supply for the resource of interest over a chosen time interval. This can then be solved using Optimal Transport. Thus, in this section we will make a discrete OT formulation incorporating the forecasted supply and demand, which then will be solved using the Sinkhorn algorithm 1. In particular, one of the resources will be used for illustrative purposes since the concept is the same for them all.

Here we will use the Haversine distance as the defined cost for transporting one tonne of goods from location x_i to y_j , and leaving abstract costs such as the effects on the public roads or the wear and tear on roads from the heavy trucks for future implementations. Now, having the supply and demand of a resource, here soil, at N future construction sites p_i , $i = 1, \dots, N$ for some time interval starting at the beginning of the first project and ending at a chosen time instant τ . Partition the construction sites into two sets; $x = \{x_i\}_{i=1}^n$ with corresponding masses $\mu = \sum_{i=1}^n \mu_i \delta_{x_i}$ and $y = \{y_j\}_{j=1}^m$ with corresponding masses $\nu = \sum_{j=1}^m \nu_j \delta_{y_j}$ representing the surplus and deficits, respectively, where $m + n = N$.

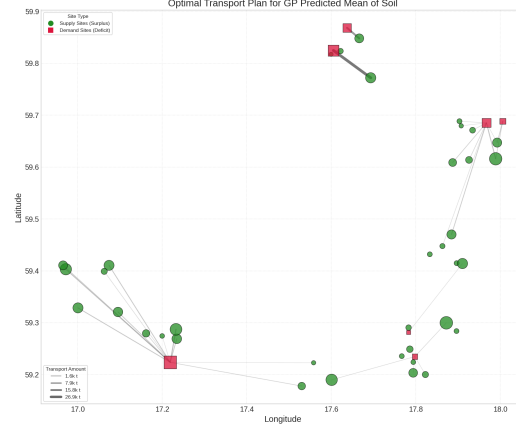
Once we have the cost matrix $\{C_{ij}\}_{i,j=1}^{n,m} = \{c(x_i, y_j)\}_{i,j=1}^{n,m} \in \mathbb{R}^{n \times m}$ we have to find the resource flow $\pi_{ij}^r = \pi^r(x_i, y_j)$ minimising the entropic UOT. Hence, we want to find dual potentials $(f, g) \in \mathbb{R}^2 \times \mathbb{R}^2$, maximising the dual of the discrete entropic UOT (3.20), which we aim to do using the Sinkhorn algorithm (1).

Finally, we use the GP predictor to forecast the soil demand and supply for the new set of projects for the time interval 2025-01-04 to 2027-07-08. Using

the forecasted measures as input to the Sinkhorn algorithm together with the cost matrix we get our sought after transport plan. This transport plan can be visualised for soil by plotting its corresponding transport map.



(a) Forecasted cumulative construction site surplus/deficit for soil for the period (2025-01-04 to 2027-07-08).



(b) Optimal Transport plan for soil based on the GP predicted mean, using the Sinkhorn algorithm.

4.5 Discussion

The primary objective of this case study has been to explore and demonstrate the potential of predictive modelling, specifically using Gaussian Processes, for forecasting resource excavation and demand in large construction projects. Furthermore, it aimed to illustrate how these predictions can be used in an Optimal Transport framework for resource allocation planning. Let us emphasize that the focus was on methodological exploration rather than seeking definitive results and high performing models on real world data. Extending this exploration to the treatment of the STA's actual data from ELSA (as well as other information sources), when the collection is mature enough, needs several adjustments and improvements. The current data generation and simulation aims to mimic real world data to some extent but it is, however, a major simplification of the dynamics and complexities that is most likely present. Such varieties can include multiple contractors with possibly different estimation error distributions, more nuanced dependencies between features or activities and a significant increase in number of

different types of construction works. The motivation of the choice of using Gaussian Processes for prediction was based on their flexibility and ability to quantify uncertainty as well as suitability to datasets leveraging feature engineering for seasonalities and trends rather than showing pure time dependency. Although, time dependent models such as state space GPs could potentially capture such temporal dependency, this study assumed independency between different construction states and their corresponding resource quantities excavated, which seems reasonable for a prototype study like this and given the current understanding of reused materials inter project-wise. Alternative machine learning methods might also be suitable for this type of data such as Gradient Boosting Machines or as data grows Neural Networks (NN) or Deep NNs. Possibly, if any underlying physics dependencies can be identified, Physics Informed NNs might be of special interest.

The subsequent application of Unbalanced Optimal Transport with entropic regularisation on the predicted means of the GP and solved via the Sinkhorn algorithm (1) is well suited because of the unbalanced nature of excavation and demand of resources over different construction sites. This could possibly be further adjusted to incorporate a more dynamic adaptability using dynamic UOT, which could account for the temporal evolution of supply, demand and costs, allowing for more adaptive and realistic transport plans over a project’s lifecycle. Furthermore, one could possibly leverage on the prediction intervals by combining Distributionally Robust Optimisation (DRO) with UOT, as in [27], providing transport plans for worst case and best case scenarios. Leveraging DRO and UOT, one can explore the suitability of further learning methods providing more robust prediction intervals (considering that CI from GP assumes the kernel reflects the true function well and that prior assumptions of noise holds), such as Conformal Prediction (CP). CP do not rely on model errors or distributions of data and can provide certain guarantees of the interval [24]. This opens for a combination of models, letting the CP wrap the underlying used model, in this case GP, and provide a robust CI to be used in Robust UOT. Moreover, there are many considerations to make when developing a cost function that is representative of the real world scenarios including gas prices, electricity prices, the effects on public roads, wear and tear on roads, availability of cargo transporters such as dumpers and cost of storing resources on-site. Therefore, possibly implementing both a stochastic cost and a dynamic time dependent cost could potentially improve the real world imitation.

Despite the simplifications made in the case study, the results suggests further explorations and implementations of predictive models for resource forecasting in large construction projects. This can likely be a valuable asset for the STA, providing nuanced data driven insights that can be leveraged by the STA during project planning as well as before negotiations. As discussed, the set of possible advancements is rich. Further kernel selection, kernel combinations, robust hyperparameter tuning and possibly use of heteroscedastic GPs using Variational Inference, if residual analysis on real data indicates heteroscedasticity, could potentially improve the GP modelling. With access to a grown ELSA data set, a more thorough analysis and optimisation could be done. Furthermore, this can likely already be made for the existing ELSA dataset consisting of activity data from smaller construction projects related to the West link.

In conclusion, by considering the possible advancements and adjustments when applied to real world data, predictive modelling will likely provide the STA with a valuable addition to their decision toolbox. In addition, by combining forecasts, predictive intervals and subsequently leverage these in a robust Optimal Transport setting, the STA can leverage a quick development of structured transport plans, which can be. Finally, it is important to acknowledge the real world complexity of logistics which might involve complexities and dynamics not captured by standard OT methods. However, it can provide initial insight, highlighting potential ways to increase efficacy, hence be a basis for more strategic actionable planning.

Chapter 5

Discussion and Conclusion

5.1 Discussion

Large infrastructure projects undertaken by the Swedish Transport Administration, such as the Cross-connection Södertörn, demands an enormous amount of transportations of resources and materials, particularly Rock, soil, gravel, cement and reinforcement bars. Not handling these massive bulk materials properly increases not only project cost but also GHG emissions, road and truck wear as well as public road disturbances. Furthermore, resources can quickly add to massive amounts if not handled in time as illustrated in figure 4.2. Thus, addressing the question of how predictive methods can be implemented in large construction frameworks, such as the Cross connection Södertörn, for forecasting of resource occurrence is an important question. Furthermore, the problem of finding a strategic transport plan for these masses is a comprehensive task, hence providing the STA with an initial transport plan carried out via Optimal Transport can also be valuable. This study addressed these questions by implementing this two-step methodology by predicting on-site excavated resource supply and landfill demand and combining this with entropically regularised unbalanced optimal transport. Furthermore, the thesis aimed to provide a breadth first introduction and exploration rather than a deep and theoretical pursuit. Therefore, much details, such as proofs of theorems used, are missing as well as a deeper discussion about the existence of an optimal transport plan to the different OT approaches. Although, there exists comprehensive literature covering

these areas, including [26, 2, 20, 6], but also [17, 16], to name a few.

This report highlighted both opportunities and challenges regarding the questions above. The GPR model turned out to be successful on the generated data using a Matérn kernel. Although, the uncertainty quantification indicated a significant uncertainty in the resulted forecast. This highlights the importance of selecting a suitable kernel and the hyperparameter tuning but also the incorporation of methods capturing the heteroscedastic nature of the data, such as using variational inference. Furthermore, it also shows the need for alternative uncertainty quantification methods such as conformal prediction. In summary, incorporating predictive methods in large construction sites turned out to be achievable. Thus, given the promising results of incorporating predictive models for bulk-mass forecasting, it is recommended the STA pursue the implementation to the real world data – adding these methods to their existing estimation toolbox. Moreover, alternative forecasting methods such as NNs or GBMs might be of interest in this regression problem as well. In particular, online learning methods could potentially be valuable in this type of settings where activity data is continuously updated.

Given the natural imbalance between demand and excavation, classic OT such as the Kantorovich formulation (3.5), and especially Monge (3.4), was not suitable to find an optimal transport plan. Thus, this study explored the possible implementation of UOT. In particular, the entropically regularised UOT, which turned out to be satisfactory for finding a transport plan in this setting. The results indicates that it is possible and that it has potential to combine the predictive means with the OT setting in a large construction framework. It further shows, that the Sinkhorn algorithm 1 is much suitable for finding a transport plan to this.

This study brought up some of the applications of OT, which is vast. In particular, the Wasserstein metric and its extensions can be applied in a number of areas. This was therefore presented as a diversion from the case study topic, to demonstrate some of the gained benefits of the application of OT to different areas, not least in the machine learning community.

There are many challenges in designing a robust OT framework for this type of problem, in particular on the uncertain outputs. Therefore, it is important to quantify the uncertainties and then handle these in a robust optimisation setting. Therefore, we suggest to further investigate the possibility to combine predictive methods, uncertainty quantification with distributionally

robust optimisation for UOT. Moreover, as expressed in the text, it is a challenge to construct a cost function reflecting the real world cost incorporating fuel prices, vehicle availability and sustainable consequences, which, hence, is a recommendation for study in future investigations. Furthermore, dynamic modelling using dynamic OT and extending the forecast to state space GP could possibly provide a more detailed output as well as capture temporal autocorrelations better in the real world data. Furthermore, as the dataset ELSA grows and if the amount of construction sites in the forecast increase, it is crucial to parallelise the computations to GPUs, which the GPflow package as well as the Sinkhorn algorithm are well suited for. Furthermore, the rather small information about external factors in the data suggests a need for extensive feature engineering for more robust forecasts in the future.

Finally, it should be emphasized that these methods cannot be a replacement of strategic transport and logistics planning. They are merely as good as the underlying data allows them to be.

5.2 Conclusion

In conclusion, this thesis have investigated how a combination of predictive models and optimal transport can be implemented to bulk-material management in large construction sites handled by the STA. The results shows that this likely can be done in a real world setting, such as in the Cross-connection Södertörn. As the ELSA database grows, this could be a promising tool, providing a quick and initial information basis to both decision makers and those responsible for strategic logistic planning. Therefore, we encourage further investigation of how to integrate predictive modelling and optimal transport in this framework. Finally, we encourage the STA to implement these methods for test and evaluation in the construction of more robust methods that have the potential to aid many entrepreneurs, logistic planners and the STA in planning as well as in negotiations.

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Appendix A

Data generation

Due to the confidentiality and current limitations of the available data in ELSA a synthetic dataset was generated in an attempt to imitate real-world, although simplified, behaviour of the resource flow in larger construction projects.

To imitate larger projects, several base stations or main projects with corresponding coordinates were selected from a list of locations, "Name": (x, y) e.g. "Trafikplats Hallunda E4": (59.25, 17.83). With these base stations as starting points project coordinates were drawn uniformly from $[-0.01, 0.01] \times [-0.015, 0.015]$ and added to the base stations coordinates.

A project's start date is determined by drawing a uniformly random integer in $\{30, \dots, 1095\}$ and subtracted from a base date. The duration of a project depends on the size of the project which are denoted and taken uniformly from $\{Small, Medium, Large, Mega\}$. For instance, the duration of a Mega-project is drawn uniformly from $\{500, \dots, 1095\}$. The total project cost is also assumed to depend on the size of the project. This cost is a proposal by a contractor and is generated as a noise factor $\varepsilon \sim \mathcal{U}(a, b)$ (at the moment we use $a = 0.85$, $b = 1.05$) multiplied with a base cost (e.g., 15 MSEK for Small, 800 MSEK for Mega) of the size of a project to account for variability in contractor proposals.

The condition of the ground for each project is also selected uniformly from a categorical list $\{Easy\ Soil, Mixed\ Ground, Clay\ Dominant, Rock\ Encountered\}$. This affects the sort of activity generated, mainly excavation of resources,

e.g., time duration, weight of excavated resources.

A representative, although simplified, sequence of activities plausible to affect resource transport and availability (e.g., site preparation, excavation, landfilling and paving) was constructed, tailored to project types such as tunnel, bridge and highway sections. For instance, a tunnel naturally implies a larger amount of excavated resources when built on land. In landfills, a resource is given a weight or probability as $P(Gravel) = 0.6$, $P(Soil) = 0.2$, $P(Sand) = 0.1$, $P(Clay) = 0.1$. As mentioned, the sequence is updated dynamically and depending on the ground condition the excavation activities are added accordingly.

The scheduling of activities are generated as they begin at a given start date and then the next activity begins at the previous activity's end date with an added uniformly distributed variable in $\{-7, -6, \dots, 6, 7\}$ (this can however perhaps be adjusted for better reflection of reality). Each activity has a base duration and depends on the size of the project and is further multiplied by a uniformly distributed factor in $[0.5, 1.5]$ accounting for some variability between different projects. Furthermore, seasons are added to account for ground frost, where during the winter (Dec, Jan, Feb) it is a 50% chance of ground frost slowing down an activity. The slowing down factor is drawn uniformly from $[1.1, 1.5]$.

Resources corresponding to an activity are quantified in tonnes i.e., actual resource excavation/landfill weight. The actual amount of resource that is excavated or used as landfill is based on the size of the project, the ground distribution and the relative weight of rock, soil, gravel, clay and sand, using soil as reference weight. Then a contractor's prior estimate of the actual value is set to be the multiplication of the actual value and a normally distributed random variable with mean $\mu = 1.05$ and standard deviation $\sigma = 0.15$ accounting for a tendency of a slight overestimate of the final value (This can be adjusted to more realistic values if there are any). This multiplicative error naturally leads to heteroscedasticity. Heteroscedasticity is reasonable since it is natural to have a larger deviation in the estimate when much larger masses and projects are considered.

Finally, there is a possibility of toxic/contaminated resources in the excavation process, excluding the possibility of reusability. Thus the contaminated resources must be transported for depot somewhere, hence they need to be accounted for. Naturally, soil is the major resource contaminated. However,

at the moment both soil, clay and rock are considered to have the possibility to be contaminated. The distribution of a resource of these to be contaminated is assumed to be Bernoulli with the probability of being contaminated is $p = 0.2$. Additionally, there might be a chance of sending non-toxic resources to depot, hence we let this be modelled Bernoulli wise as well with $p = 0.05$.

